

Congressional Redistricting - Command Reference

Run `setup_env.bat` first to enable shortcuts
Updated: 2026-01-18

Quick Project Shortcuts

Pipeline:

run
runtest
cancel
validate

Testing:

test
unittest
inttest
e2etest

Dashboard:

dash
gendash
master

Data:

dl
dlcheck
dlval
dldemo
buildadj

Pipeline

Common:

```
run -v v1
# All years (2-4h)
run -y 2020 -v v1
# Single year (~1h)
run -y 2020 -v v1 -st CA TX
# Specific states
run -v v1 -r
# Fresh run
run -v v1 -s nation
# National only (fast)
run -w 12 -v v1
# Custom workers
```

Test/Debug:

```
runtest -y 2020 -v test
# Test run (outputs/dev/)
runtest -y 2020 -v t -st VT
# Single state test
run -p -v preview
# Dry run
run -v v1 -d
# Debug mode
```

Experimental:

```
run -pm unweighted -v v1
run -rt experiment
```

```
--experiment-name test123
-v v1
```

Tests

```
test           # All (18s)
unittest       # Unit tests
inttest        # Integration
e2etest        # End-to-end
pytest tests/ -x # Stop first
pytest tests/ --cov # Coverage
```

Dashboard

```
dash --year 2020 --version v1
gendash --year 2020 --version v1
master
start outputs/index.html
```

Data Management

Download Orchestrator (NEW - Parallel):

```
dl
# Check all data (default: all stages,
# all years, check-only)
dl --stages redistricting --year 2020
# Download redistricting data
dl --stages redistricting demographics \
--year all --workers 12
# Multi-year parallel (4-8x faster)
dl --stages metros enacted_districts \
--year 2020
# Download metros and CDs
dl --stages redistricting --year 2020 \
--force
# Force redownload
dlval --year all --create-markers
# Mark complete data
```

Available Stages:

- redistricting - Census tracts/blocks + pop
- demographics - Age, race, sex data

- elections - Presidential results
- places - City/town boundaries
- metros - CBSA/Metro boundaries
- enacted_districts - Baseline CDs

Legacy Downloads (Individual):

```
dldemo -y 2020
# Demographics only
diselect -y 2020
# Elections only
dlplaces -y 2020
# Places only
```

Processing:

```
buildadj -y 2020
# Build adjacency graphs
checkconn -y 2020
# Check connectivity
validate -y 2020 -v v1
# Validate outputs
```

Emergency

```
cancel # Kill all Python
```

Claude Skills

```
/run-redistricting
/run-tests
/debug-tests
/debug-pipeline
/enhancement-plan
/enhancement-implement
/create-state-map
/create-national-map
/generate-dashboard
/data-validate
```

Pipeline Flags

Short Flags (all have long equivalents):

Short	Long	Values	Default
-h	--help	-	-
-y	--year	2020/2010/2000/all	all
-v	--version	string	v1
-s	--stages	data/states/nation	all
-st	--states	codes	all
-w	--workers	1-24	12
-r	--reset	flag	-
-p	--print-only	flag	-
-d	--debug	flag	-
-ey	--election-year	2020/2016	2020
-pm	--partition-mode	edge/unweighted	edge
-rt	--run-type	prod/test/exp	prod

Long-only Flags:

- --dpi N - Quality (def: 150)
- --minimum-boundary-length N - Filter adjacencies (def: 10m)
- --skip-political - No politics
- --skip-demographic - No demos
- --reprocess - Redo all states
- --experiment-name - Experiment name

Stage Examples:

- -s data - Data only
- -s states nation - Skip data
- -s nation - Post-processing only

Download Orchestrator Flags:

- --stages - Stages to download (redistricting, demographics, elections, places, metros, enacted_districts)

- --year - Census year (2000/2010/2020/all)
- --workers N - Parallel workers (def: 4)
- --states - Specific states (e.g., CA TX NY)
- --check-only - Check cache, no download
- --version - Force redownload
- --output-dir - Output directory
- --validate - Validate Downloads:
 - --year - Year to validate (2000/2010/2020/all)
 - --stages - Stages to check
 - --create-markers - Create completion markers
- --check-only - Check without modifying

Tests:

- -v - Verbose
- -s - Show prints
- -x - Stop first fail
- --cov - Coverage
- -k PATTERN - Filter

Performance

DPI: 100 (dev) — 150 (default) — 200 — 300 (print)

Times:

- Multi-year: 2-4h
- Single year: 1h
- Small state: 30s-2min
- Large state: 3-5min
- Tests: 18s